Investigation: FORJAS TAURUS SA (CVM code 6173), fiscal year 2012, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2012 DFP, version 3, filed on 2013-04-24; origin: ENET; type: consolidated; original scale: —.
Versions known on the date: 1 (2013-03-26); 2 (2013-04-04); 3 (2013-04-24).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=26284
- Structure read from a document filed after the date (study 1 behaviour, amendment 8)
- ENET version: the cache keeps values already in reais and did not record the page's scale (fetches before 2026-11-09)

Possible manipulation (engine v1, study 1)

Beneish M-score
  Value: −2.002 (index)
  Alert by the study 1 rule: no
  Rule: M > −1.78 (pre-registration, section 4)
  Percentile in sector and year: 91 (n = 16)
  Percentile in group and year: 81 (n = 334)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

DSRI, days' sales in receivables index
  Value: 0.931 (index)
  M-score component; it has no rule of its own
  Percentile in sector and year: 17 (n = 21)
  Percentile in group and year: 33 (n = 412)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

GMI, gross margin index
  Value: 1.096 (index)
  M-score component; it has no rule of its own
  Percentile in sector and year: 55 (n = 21)
  Percentile in group and year: 69 (n = 426)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

AQI, asset quality index
  Value: 2.039 (index)
  M-score component; it has no rule of its own
  Percentile in sector and year: 88 (n = 21)
  Percentile in group and year: 97 (n = 421)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

SGI, sales growth index
  Value: 1.134 (index)

M-score component; it has no rule of its own
Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 55 (n = 427)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.005 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 53 (n = 16)
Percentile in group and year: 65 (n = 351)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.839 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 17 (n = 21)
Percentile in group and year: 19 (n = 420)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.015 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 69 (n = 21)
Percentile in group and year: 65 (n = 438)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.027 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 55 (n = 21)
Percentile in group and year: 53 (n = 432)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 2.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 27.2 days)
Percentile in sector and year: 68 (n = 20)
Percentile in group and year: 62 (n = 396)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 15.44 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 69 (n = 21)

Percentile in group and year: 75 (n = 394)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.021 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 69 (n = 21)
Percentile in group and year: 64 (n = 437)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 2.919 (index)
Components: X1 = 0.233; X2 = 0.110; X3 = 0.091; X4 = 0.400 (indices)
Percentile in sector and year: 75 (n = 22)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 5/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-30
- DFP 2010, v2: 2011-03-30
- DFP 2010, v3: 2011-04-12
- FCA 2012, v1: 2012-02-29
- DFP 2011, v1: 2012-03-27
- FCA 2012, v2: 2012-05-18

- FCA 2012, v3: 2012-06-11
- DFP 2012, v1: 2013-03-26
- DFP 2012, v2: 2013-04-04
- DFP 2012, v3: 2013-04-24

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

the open-data opinions section is only for the latest DFP version; the version used came from ENET

Auditor in the 2010 FCA: KPMG Auditores Independentes.

Auditor in the 2011 FCA: KPMG Auditores Independentes.

Auditor in the 2012 FCA: Ernst & Young Terco Auditores Independentes S/S, KPMG Auditores Independentes.

Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: entry of Ernst & Young Terco Auditores Independentes S/S, with no other firm leaving.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Structure read from a document filed after the date (study 1 behaviour, amendment 8)
- ENET version: the cache keeps values already in reais and did not record the page's scale (fetches before 2026-11-09)
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-11T00:42:24+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: TAURUS ARMAS S.A..